Data Backup & Analysis Guide

Step 0: Stimulus Backup

Back up stimulus files to hard disk and Nextcloud, Labguru (small stim only<100Mo).

- Stimulus files (.vec/.bin) : during exp directly on labguru, later on nextcloud and hardrive. Include the date in the filenames.
- Generating code (at the date you used it to avoid version issues) : nextcloud and hardrive

Notes :

- (.vec/.bin) for shared stimuli can be stored in shared folder in nextcloud
- In Nextcloud use the Team Marre folder, then choose your PI folder, then Data

Step 1: Backup Log

Save the experimental log in Labguru. It can also be good to have that information in a .txt or .pdf file in the experiments folder (nextcloud, hard disk). One option is to save the Labguru page as a pdf (print page to pdf option in your browser). The following informations are useful :

- List of stimuli and parameters
- Rig ID (and MEA parameters)
- Position in the retina
- Illumination
- Filter used
- Experiment date and session information
- Animal/subject identifier
- Experimenter name
- Special conditions or observations
- Link to lab notebook or experiment log
- List of found cell types

Step 2: Backup MCD Files

Save MCD to hard disk (to keep not sorted data).

Step 3: Automatic Sorting

Run automatic sorting (spikingcircus)

Step 4: Manual Sorting & Pipeline

Run manual sorting and first analysis pipeline (Notebook 1 that deals with the sorting)

- Complete manual sorting (phy)
- Run analysis pipeline

Step 5: Backup Sorting Folder

Save the Sorting folder to hard drive.

(! / We need to make sure the Sorting properly created by the pipeline)

Step 6: Run the rest of the pipeline analysis

Notebook 2+. You can modify the analysis pipeline to your need, we'll save it just after

Step 7: Backup Analysis + Pipeline Folder

Save Analysis output folder and Pipeline code folder to Nextcloud and hard disk.

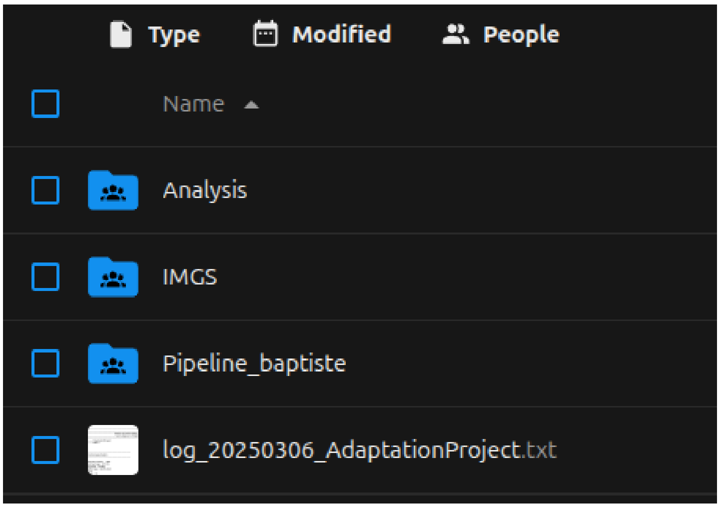
Step 8: Back Up Back Up

Copy the content of the hard drive to a second hard drive. That way when you remove the experiment from your computer, you will still have two copies.

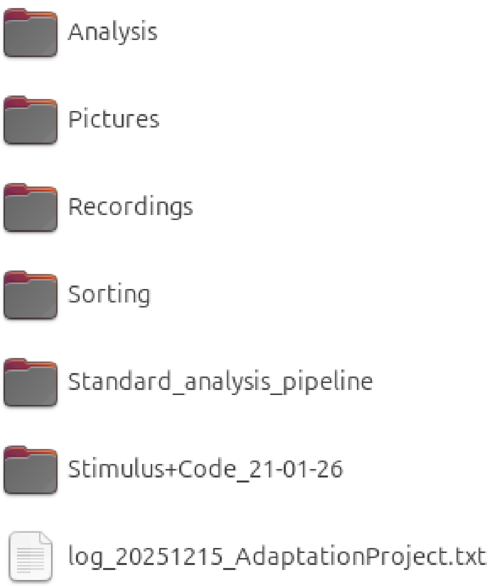
Bonus Step : Further Analysis

If you do successive analyses of an experiment, try to save your analysis results on the first hard drive / nextcloud from time to time.

Example folder structure on Nextcloud



Example on Hard disk



Additional Documentation

Hard Drive Formatting

Any new disk must be formatted in **exFAT** to be more easily used across devices.
(If really needed, you can use NTFS. Avoid FAT32 at all costs.)
(On Ubuntu you may need to install the following packages to use exFAT

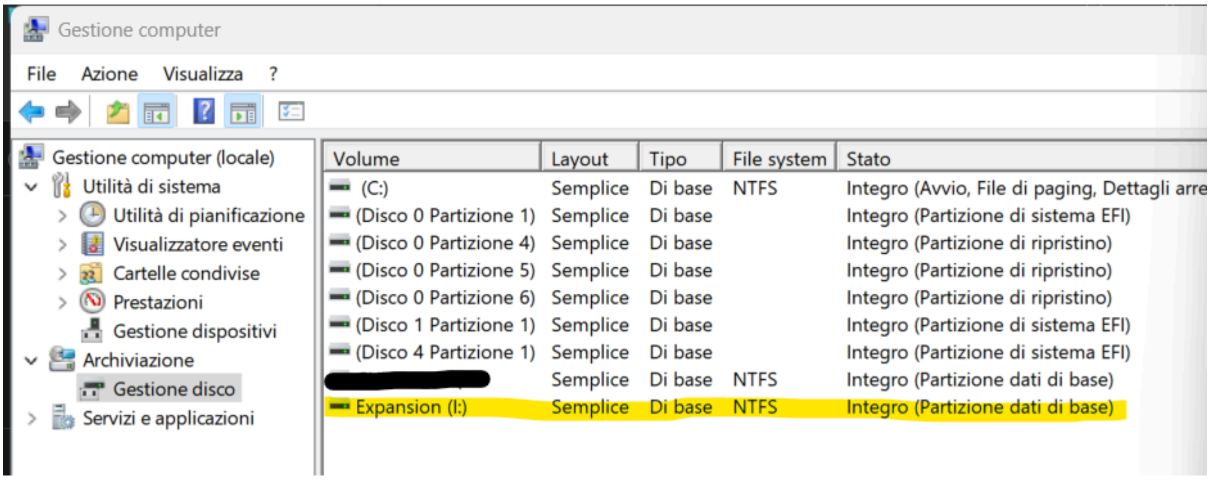
- sudo apt install exfatprogs
- sudo apt install ntfsprogs

)

0. Connect the drive
NB: if not brand new - Backup somewhere else the content of the drive you want to keep and clean the drive (This is to avoid doing file compression, which takes a long time and do the formatting quickly without files)

(Windows)

1. (optional) Rename the expansion: This PC → right-click on the expansion → rename
2. Right-click on *This PC* → (more options) → Manage disk → (Storage) Manage disk (it can take some time to display the units) (or Wnds + R → diskmgmt.msc)
3. Right-click on the expansion to format → Format → Select as file system **exFAT** and **execute fast formatting**
4. Once the file system is updated, it is done and you can close.



(Linux)

1. Open the Disks utilitary nby searching ‘Disks’ in Ubuntu research bar
2. Select chosen disk on the left
3. In the top section, left to your disk name click the 3 dots than Format Disk
4. Select **exFAT** format and perform formatting

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